In [ ]:

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# EXPERIMENT NO. 4: Advanced Data Visualization using Matplotlib and Seaborn
# Step 2: Import Required Libraries and Load Dataset   Pragya Singhal CU24220050
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Set visual style theme  Pragya Singhal CU24220050 , 43
sns.set_theme(style="whitegrid")
plt.rcParams['figure.figsize'] = (10, 6)

# Load your COVID-19 dataset
df = pd.read_csv("covid_19_clean_complete.csv")

print("--- DATASET HEAD ---")
print(df.head(), "\n")


# Step 3: Bar Chart Pragya Singhal CU24220050 , 43
plt.figure(figsize=(12, 6))
top_countries = df.groupby('Country/Region')['Confirmed'].sum().nlargest(10).res

sns.barplot(data=top_countries, x='Confirmed', y='Country/Region', palette='Blue
plt.title("Top 10 Countries by Total Confirmed COVID-19 Cases", fontsize=14, fon
plt.xlabel("Total Confirmed Cases", fontsize=12)
plt.ylabel("Country", fontsize=12)
plt.tight_layout()
plt.show()


# Step 4: Line Chart Pragya Singhal CU24220050 , 43
if 'Date' in df.columns:
    df['Date'] = pd.to_datetime(df['Date'])
    daily_trend = df.groupby('Date')['Confirmed'].sum().reset_index()

    plt.figure(figsize=(14, 6))
    sns.lineplot(data=daily_trend, x='Date', y='Confirmed', color='crimson', lin
    plt.title("Global COVID-19 Confirmed Cases Trend Over Time", fontsize=14, fo
    plt.xlabel("Date", fontsize=12)
    plt.ylabel("Confirmed Cases", fontsize=12)
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.show()


# Step 5: Histogram Pragya Singhal CU24220050 , 43
plt.figure(figsize=(10, 6))
sns.histplot(df['Deaths'], bins=30, kde=True, color='purple')
plt.title("Distribution of COVID-19 Deaths Across Regions", fontsize=14, fontwei
plt.xlabel("Deaths", fontsize=12)
plt.ylabel("Frequency", fontsize=12)
plt.tight_layout()
plt.show()


# Step 6: Box Plot Pragya Singhal CU24220050 , 43
plt.figure(figsize=(8, 6))
sns.boxplot(y=df['Active'], color='orange', fliersize=5)
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plt.title("Box Plot of Active COVID-19 Cases (Outlier Detection)", fontsize=14,
plt.ylabel("Active Cases", fontsize=12)
plt.tight_layout()
plt.show()


# Step 7: Scatter Plot Pragya Singhal CU24220050 , 43
plt.figure(figsize=(10, 6))
sns.scatterplot(data=df, x='Confirmed', y='Deaths', alpha=0.7, color='teal')
plt.title("Scatter Plot: Total Confirmed Cases vs. Total Deaths", fontsize=14, f
plt.xlabel("Confirmed Cases", fontsize=12)
plt.ylabel("Deaths", fontsize=12)
plt.tight_layout()
plt.show()


# Step 8: Correlation Heatmap (Relationships among numerical features)
plt.figure(figsize=(8, 6))
numerical_df = df.select_dtypes(include=[np.number])
corr_matrix = numerical_df.corr()

sns.heatmap(corr_matrix, annot=True, cmap="coolwarm", fmt=".2f", linewidths=0.5)
plt.title("Correlation Matrix Heatmap of COVID-19 Metrics", fontsize=14, fontwei
plt.tight_layout()
plt.show()


# Step 10: Summary of Insights    Pragya Singhal CU24220050 , 43
print("--- 10. SUMMARY OF GRAPHICAL INSIGHTS ---")
print("1. Bar Chart Analysis: Highlighted the top regions carrying the heaviest
print("2. Line Chart Analysis: Showcased wave spikes and chronological spread pa
print("3. Histogram & Box Plot: Proved high skewness in metrics with significant
print("4. Scatter Plot & Heatmap: Demonstrated strong positive correlations betw
```

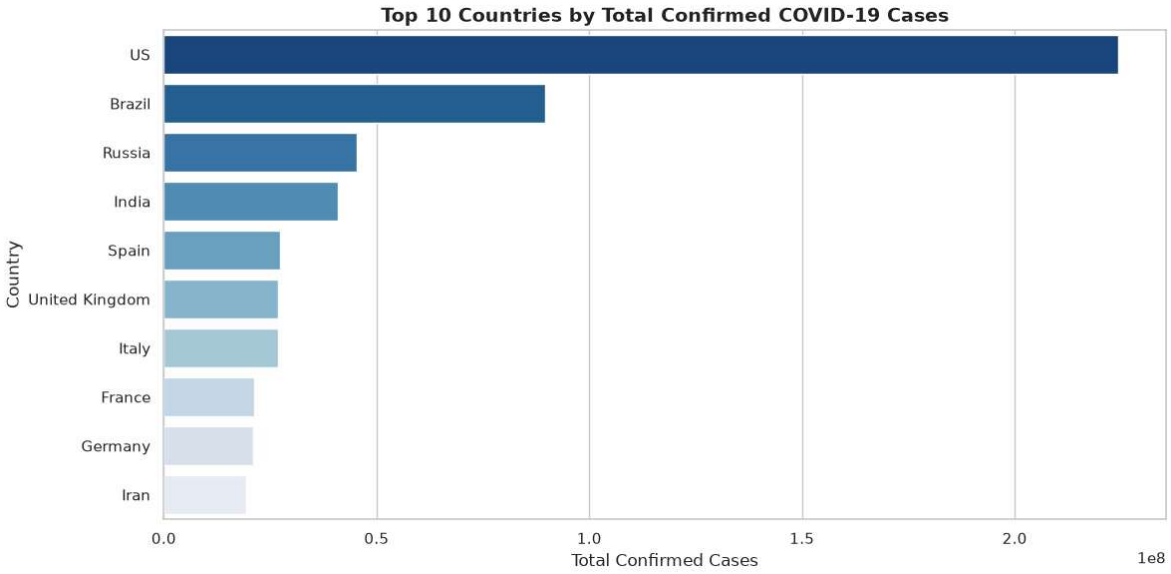
```
--- DATASET HEAD ---
  Province/State Country/Region       Lat       Long        Date  Confirmed  \
0            NaN    Afghanistan  33.93911  67.709953  2020-01-22          0
1            NaN        Albania  41.15330  20.168300  2020-01-22          0
2            NaN        Algeria  28.03390   1.659600  2020-01-22          0
3            NaN        Andorra  42.50630   1.521800  2020-01-22          0
4            NaN         Angola -11.20270  17.873900  2020-01-22          0

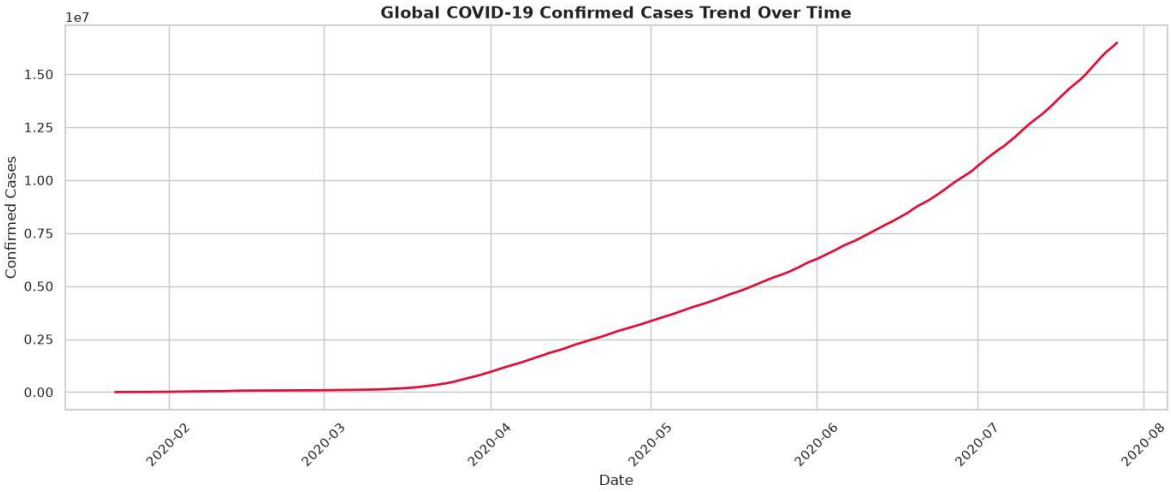
   Deaths  Recovered  Active             WHO Region
0       0          0       0  Eastern Mediterranean
1       0          0       0                 Europe
2       0          0       0                 Africa
3       0          0       0                 Europe
4       0          0       0                 Africa
```

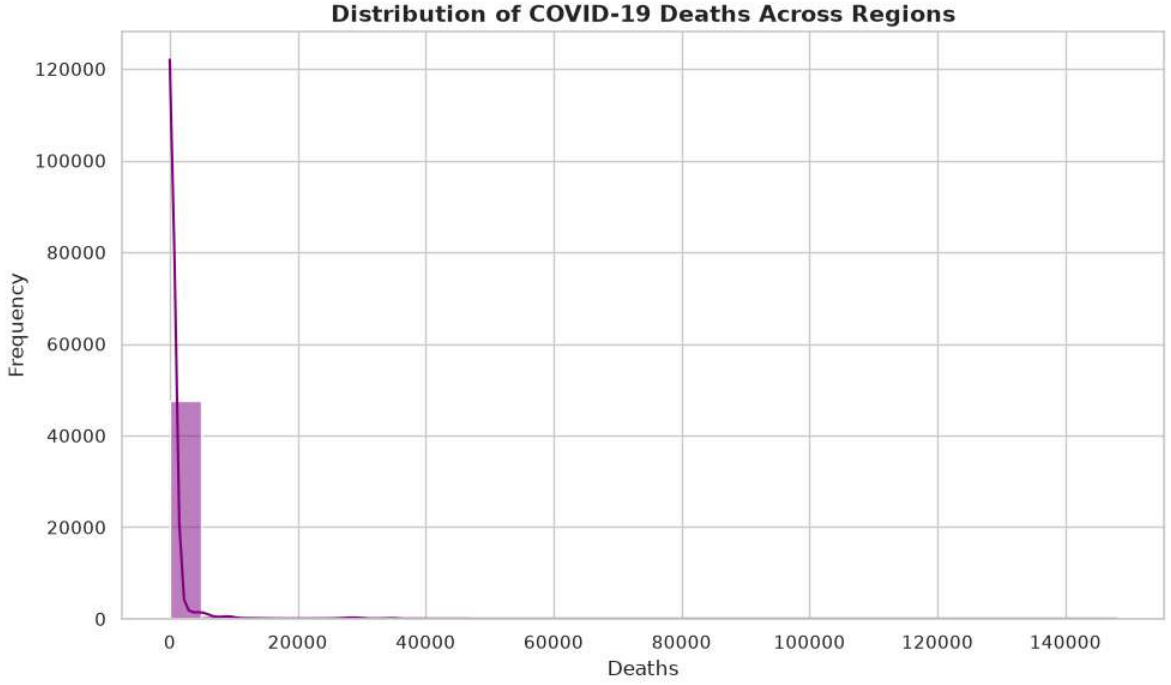
```
/tmp/ipykernel_10382/493228114.py:27: FutureWarning:

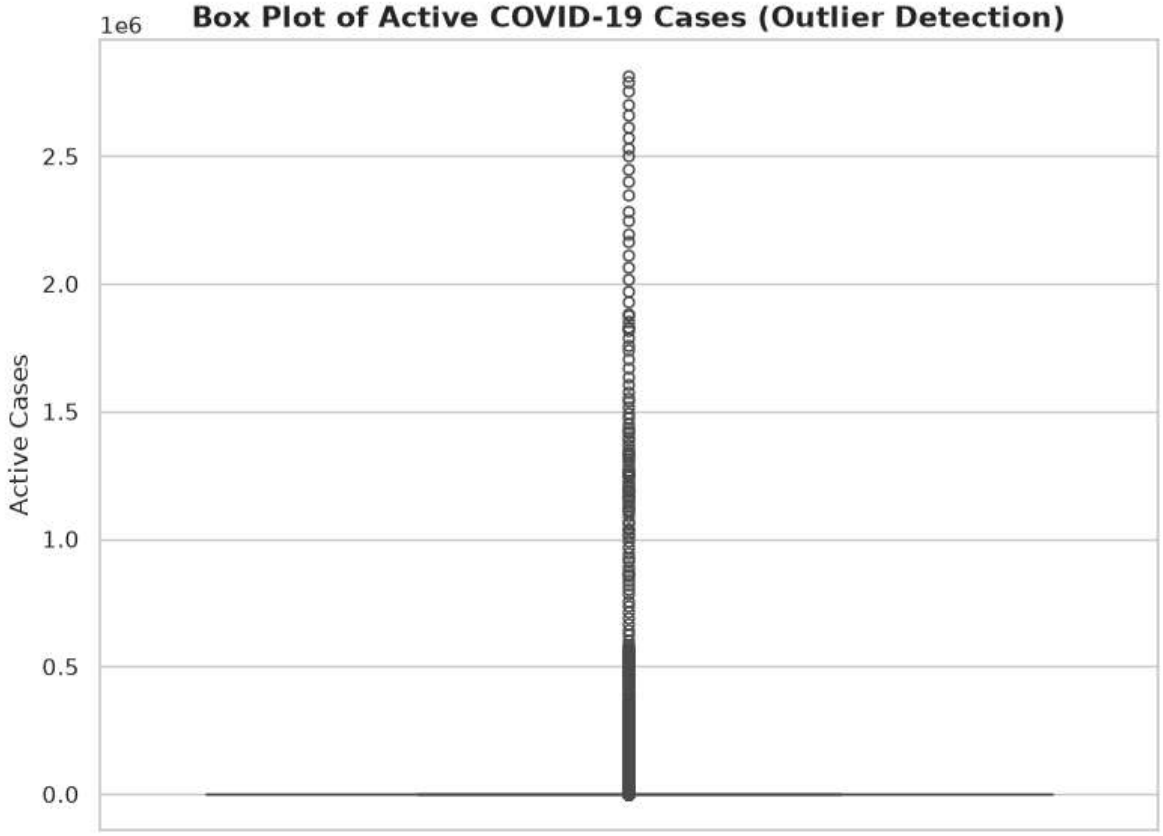
Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `y` variable to `hue` and set `legend=False` for the same effe
ct.

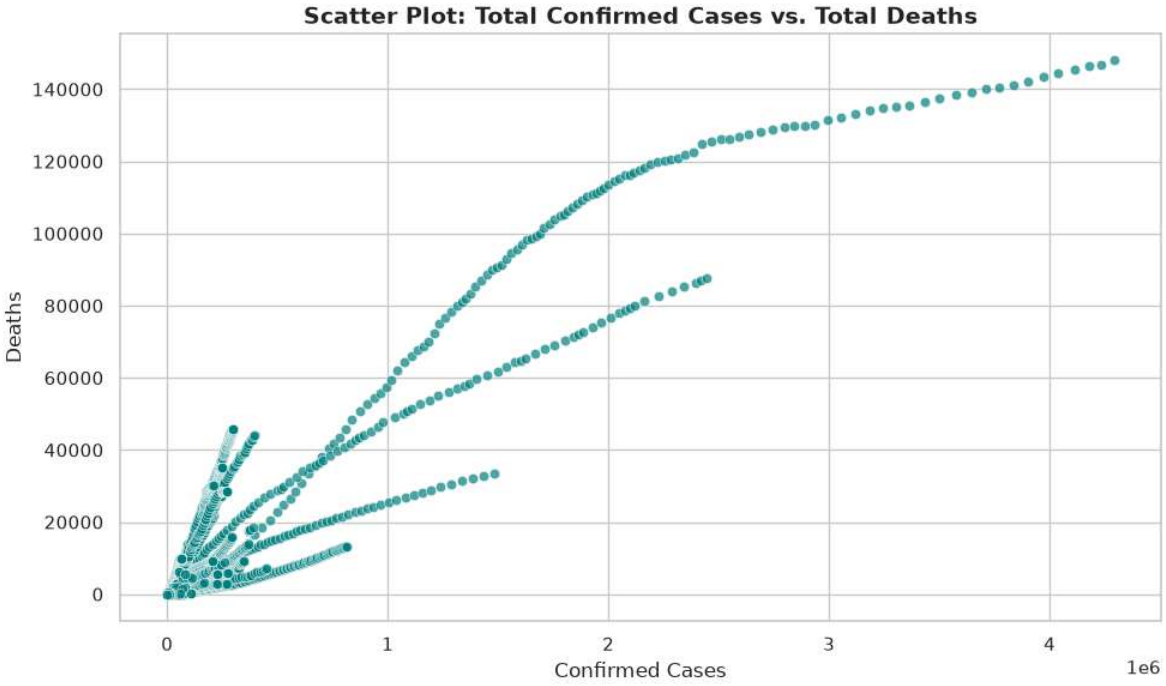
  sns.barplot(data=top_countries, x='Confirmed', y='Country/Region', palette='Blu
es_r')
```

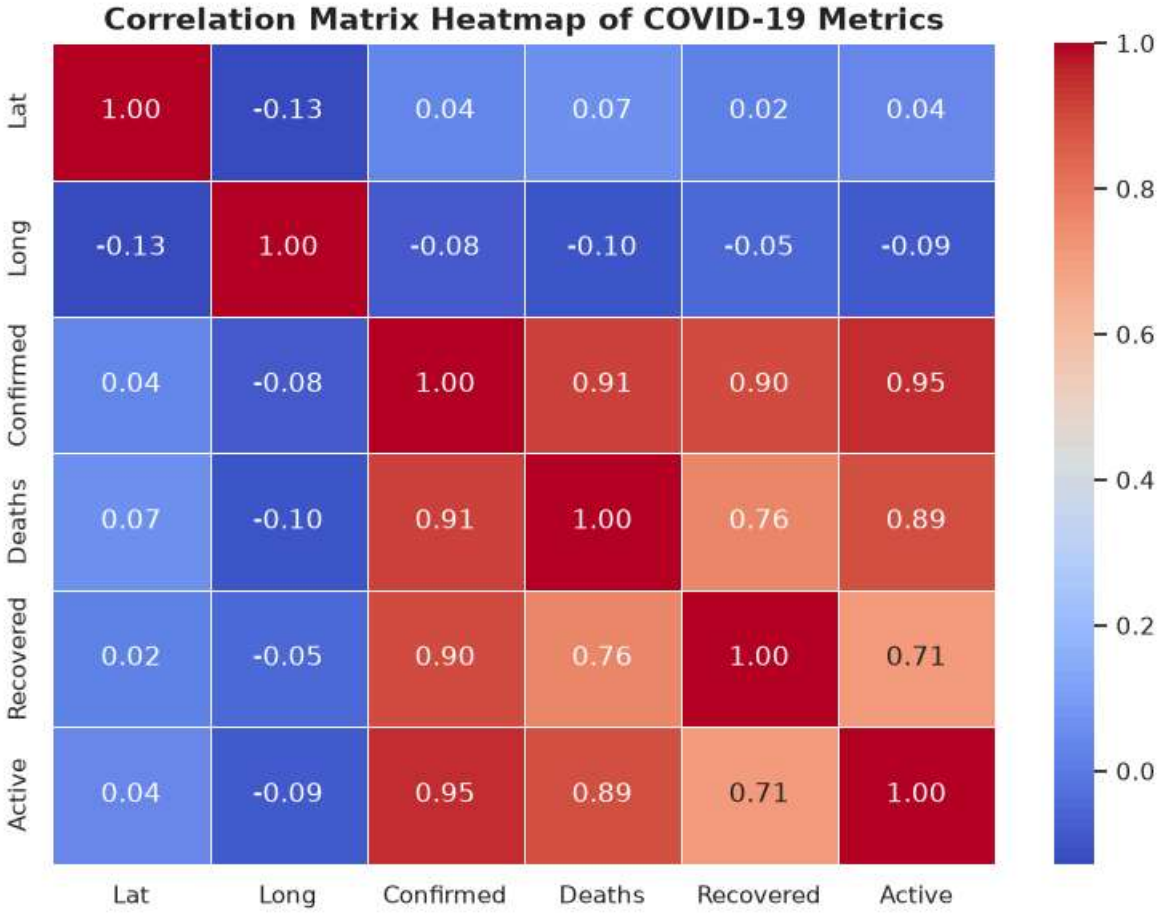

Top 10 Countries by Total Confirmed COVID-19 Cases
US
Brazil
Russia
India
Spain
United Kingdom
Italy
France
Germany
Iran
Country
0.0
0.5
1.0
1.5
2.0
Total Confirmed Cases
1e8


Global COVID-19 Confirmed Cases Trend Over Time
1e7
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Confirmed Cases
2020-02
2020-03
2020-04
2020-05
2020-06
2020-07
2020-08
Date


Distribution of COVID-19 Deaths Across Regions
120000
100000
80000
60000
40000
20000
0
Frequency
0
20000
40000
60000
80000
100000
120000
140000
Deaths


Box Plot of Active COVID-19 Cases (Outlier Detection)
1e6
Active Cases
2.5
2.0
1.5
1.0
0.5
0.0


Scatter Plot: Total Confirmed Cases vs. Total Deaths
Deaths
140000
120000
100000
80000
60000
40000
20000
0
0
1
2
3
4
Confirmed Cases
1e6



--- 10. SUMMARY OF GRAPHICAL INSIGHTS ---
1. Bar Chart Analysis: Highlighted the top regions carrying the heaviest infectio n burdens.
2. Line Chart Analysis: Showcased wave spikes and chronological spread patterns o ver time.
3. Histogram & Box Plot: Proved high skewness in metrics with significant statist ical outliers (severe hit zones).
4. Scatter Plot & Heatmap: Demonstrated strong positive correlations between tota l confirmed cases and total fatalities.